intermediate-elective

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# visualise missing data
library(naniar)

# arrange plots
library(patchwork)

# extra packages required for visualisation
library(ggtext)
library(showtext)
```

Data

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

```r
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```r
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
```

```
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

```
# new object created from aq_ins_clean
logsalinity2 <- aq_ins_clean |>
  # log + 1 transform mean abundance/liter
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(logsalinity2)
```

```
# A tibble: 1 x 24
  date_site      date                site  taxon   ave_lit med_ph med_sal med_do
  <chr>          <dttm>              <chr> <chr>     <dbl>  <dbl>   <dbl>  <dbl>
1 2023-11-14_NPB 2023-11-14 00:00:00 NPB   anneli~       0     NA      NA     NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

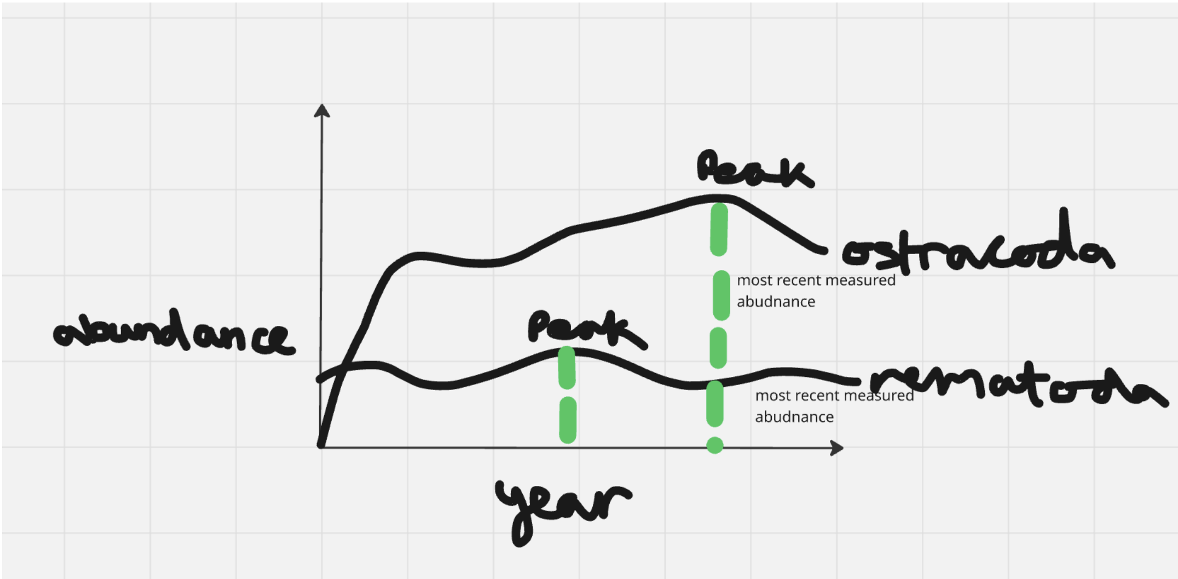
1. Explain your inspiration

Which visualization you chose for your inspiration? - I chose Steven Ponce's "Coal falls. Renewables rise."

Why it makes sense for your dataset of choice? - It makes sense as it shows how the abundance of 2 different aquatic invertebrates change through time.

which variables from your dataset will be shown in your visualization, and what visual components they map onto (axes, shapes, colors, etc.) - x-axis will be the date (year) - y-axis will be the abundance (log transformed average abundance per liter) - I will use points and lines (line graph) to show how the 2 taxons increase or decrease in abundance over time. I am choosing the taxon ostracoda and nematoda - I will also show when their abundance peaks _ I will use the same colours as the inspiration

2. Plan your figure



3. Code your figure

```
taxon_abundance <- logsalinity2 |>
  mutate(
    date = ymd(date),
    year = year(date)
  ) |>
  filter(
    taxon %in% c("ostracod", "nematode"),
    year >= 2020,
    year <= 2024
  ) |>
  group_by(year, taxon) |>
  summarise(
    abundance = mean(log_ave_lit, na.rm = TRUE),
    .groups = "drop"
  ) |>
  mutate(
    taxon = factor(taxon, levels = c("ostracod", "nematode"))
  )
```

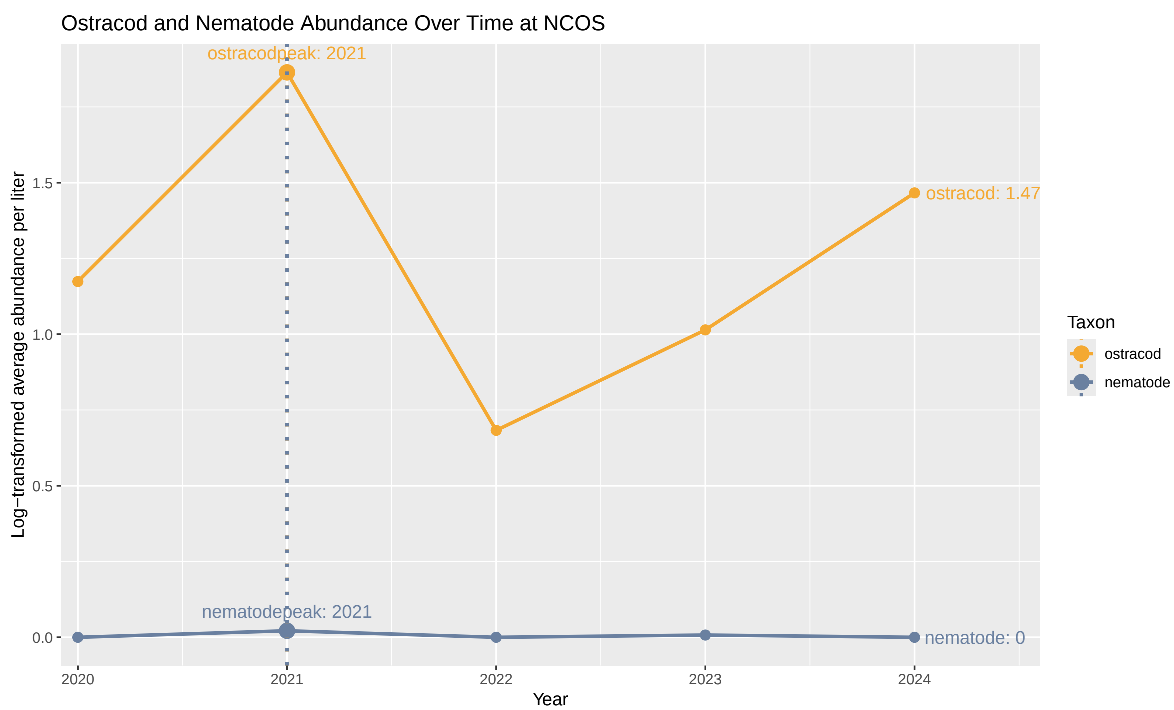
```r
taxon_peaks <- taxon_abundance |>
  group_by(taxon) |>
  slice_max(abundance, n = 1, with_ties = FALSE) |>
  ungroup()
```

```r
taxon_end <- taxon_abundance |>
  group_by(taxon) |>
  slice_max(year, n = 1, with_ties = FALSE) |>
  ungroup()
```

```r
# selecting colours for ostracods and nematodes
ostracod_gold <- "#F4A932"
nematoda_blue <- "#6A80A0"

# plot figure using taxon_abundance dataset
ggplot(taxon_abundance,
       aes(x = year,
           y = abundance,
           color = taxon,
           group = taxon)) +
  # line to connect all points
  geom_line(linewidth = 1) +
  # adding point to figure
  geom_point(size = 2.5) +
  # increasing size of peak points
 geom_point(data = taxon_peaks,
             aes(x = year, y = abundance),
             size = 4) +
  # adding dotted line for peaks
 geom_vline(data = taxon_peaks,
             aes(xintercept = year,
                 color = taxon),
             linetype = "dotted",
             linewidth = 1) +
  # labelling the peaks
  geom_text(data = taxon_peaks,
            aes(label = paste0(taxon, "peak: ", year)),
            vjust = -1,
            size = 4,
            show.legend = FALSE) +
  # labelling the end points
 geom_text(data = taxon_end,
```

```
            aes(label = paste0(taxon, ": ", round(abundance, 2))),
            hjust = -0.1,
            size = 4,
            show.legend = FALSE) +
  # adding a different colour for each taxon
  scale_color_manual(values = c(
    "ostracod" = ostracod_gold,
    "nematode" = nematoda_blue
  )) +
  # y-axis and x-axis labels and titling
  labs(
    title = "Ostracod and Nematode Abundance Over Time at NCOS",
    x = "Year",
    y = "Log-transformed average abundance per liter",
    color = "Taxon"
  ) +
# expanding x-axis so all labels fit
scale_x_continuous(expand = expansion(mult = c(0.02, 0.15)))
```



4. Describe your process

Your process of using someone else's code to create a visualization - It was honestly extremely difficult trying to use someone else's code to make a visualisation. - At first, I tried copying and pasting their code and editing it referencing my dataset. - That attempt did not work as I encountered many different errors. - I then had to use their figure as inspiration and using different code chunks that we have encountered in class as well as new lines of code (for example, geom_vline, geom_text).

Challenges you encountered as you made your visualization _ I'm not even sure if I have completed this elective how it was intended to. - The packages used in the inspiration's code were not loading on my RStudio (with errors saying my packages were outdated) - When trying to sort the figure by year, it was also quite difficult as it was the wrong list type. - After these errors, I decided to change the approach I was using.

How your visualization differs from visualizations you made in class, for assignments, or for a previous elective? - My visualisation contains very similar components as the ones we have created from class. However, it is different in that it has an added dotted line showing where the abundance peaks. Additionally, the point for peak was also increased in size - which I don't believe we have done in class. I also don't think visualisations from class included a label and end-point value.